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The risk of developing Periodontitis; A Random Forest based algorithm

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Abstract

Periodontitis is a gum disease that is the result of infections and inflammation of the gums and the bone which surround and support the teeth. In the most severe cases, teeth may loosen or fall out.
Using the ensemble machine learning method Random Forest we predict the risk of developing Periodontitis with three different models and also investigate which predictors are the most useful for the predictions. Each model is based on the same data with both systemic and local predictors but the predictors included in each model differ. The model's performances are evaluated based on different types of accuracy measures and the predictors are evaluated using Variable Importance plots and measures. The model with the most predictors included performed best, as a result of it using the most local predictors which seemed to have a higher importance in predicting, compared to the systemic predictors.

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1 Introduction

1.1 Background

Periodontitis is a type of gum disease that mainly is the result of infections and inflammation of the gums and the bone which surround and support the teeth. Periodontitis may lead to the gums pulling away from the tooth, bone can be lost and in the most severe cases the teeth may loosen or fall out. Periodontal disease is mostly seen in adults and is among tooth decay the biggest threats to dental health.

It is caused by bacteria in the mouth that infects the tissue surrounding the tooth causing inflammation around the tooth which leads to the disease. If a bacteria stays on the teeth for too long, they form a film called plaque, the plaque eventually hardens to tartar. This can then spread below the gum line which makes it harder to clean. If so, only a dental health care professional is able to remove the tartar and stop the process of the Periodontal disease.

Some of the typical warning signs are bad breath, red or swollen gums, tender or bleeding gums, painful chewing, loose teeth and sensitive teeth. Certain factors might also increase the risk of the disease such as smoking, diabetes, poor oral hygiene, stress, heredity, crooked teeth, fillings that have become defective, medication that cause dry mouth and female hormone changes. To prevent getting Periodontitis good oral hygiene is recommended and a regular professional cleaning. In more severe forms of the disease, treatments include deep cleaning of the tooth root surfaces below the gums, medications prescribed to take by mouth or to be placed directly under the gums and in some cases corrective surgery.

It is important to study Periodontal disease as it is a common disease that can lead to problematic and severe health complications. Also, it is important to understand which patients are more likely and how to predict which patients that will get the disease in the future so that actions can be made against the disease in the early stages.

Overall, Periodontitis is a serious oral health condition that has to be properly diagnosed and treated. By understanding its causes and risk factors, and developing predictive strategies, we can work to reduce its spread and impact on the public health (Centers for Disease Control and Prevention, 2023-04-26).

In this report, an algorithm for predicting the risk of developing getting Periodontitis is made using a machine-learning based method called Random Forest. Three different models, with three different setups of variables, will be evaluated to get an understanding of which variables may cause the disease. The outcome variable included in the models can categorize into one of three values. The predictors are separated into either systemic or local. Systemic refers to non tooth-specific factors, such as age, smoking habits and other or earlier diseases. Local predictors on the other hand refers directly to the teeth health. For the models that are used, the data is split up in a training set and a validation set. The training set includes 70% of the data and the validation set 30% of the data.

1.2 Purpose & Question of interest

The purpose of this report is to create an algorithm that can predict the risk of someone developing Periodontitis. This will be done by comparing three Random Forest models with different setups of predictors that categorizes into either systemic or local factors and then choosing the best of the models. So the question of interest leads to:
"How do systemic and local predictors affect the risk of developing Periodontitis according to the machine learning-based method Random Forest?".

1.3 Disposition

The main part of the report will in section 2 describe the data being used, mostly about the data sets and missing values. In section 3 you can get an understanding of the tools being used to get to the results in the report. Section 4 contains the results from the three Random Forest models and the discussion about the results and the report in general can be read about in section 6. Section 7 is the conclusion part.

2 Data

2.1 Data set

The data in the report origins from two separate data sets. One contains information about the patients, which refers to systemic predictors. This data set contains 213 observations, where each observation is a patient. The other data set contains information about local predictors, with other words the patients teeth. Every observation in this data set is a specific tooth and every patient has information about 16 teeth. So the number of observations in this data set, 3408, is simply the product of the number of patients and number of teeth per person. This means that the data are hierarchical, since each patient has a number of teeth that likely are not independent. The machine learning method in question, Random Forest, can work with data that has a hierarchical structure.

The information about the patients has been gathered during two different occasions. During the first occasion information about the majority of the predictors were gathered. During the second occasion which was 4 years later, the outcome variable and the remaining predictors were collected and measured which means that the study has a time horizon of 4 years.

2.1.1 Description of variables

As mentioned earlier the three models contain different setups of the systemic and local predictors and with different amounts of them. To get an understanding what the variables mean, look at Table 1 and Table 2 and concerning what setup every model has, look at Table 3 below.

Table 1: Description of systemic predictors

Variable	**Variable name**	**Description**
Age	age	age in years
Family History	dtmother	1 if mother suffers from periodontitis
		0 if not
		2 if patient does not know
	dtfather	1 if father suffers from periodontitis
		0 if not
		2 if patient does not know
Socioeconomic factors	dtsocec	0 if none
		1 if negative stress
		2 if poor economy
		3 if both
Patient disease awareness	dtintdis	0 if none
		1 if little
		2 if high
Current Smoking	skhab	0 if no smoking
		1 if <10 cigarettes per day
		2 if 10-20 cigarettes per day
		3 if >20 cigarettes per day
Dentist experience	opexp	1 if some (general clinics)
		2 if extensive (special clinics)
Systemic diseases	depreg	1 if pregnancy
		0 otherwise
	diab_notcontr	1 if diabetes mellitus, not controlled
		0 otherwise
	diab_contr	1 if diabetes mellitus, well controlled
		0 otherwise
	blood	1 if blood or immune diseases
		0 otherwise
	monogen	1 if monogenetic diseases disturbances
		0 otherwise
	granulom	1 if granulomatous diseases
		0 otherwise
	drug	1 if drug induced parodontal disease
		0 otherwise
	osteop	1 if osteoporosis
		0 otherwise
	malnutr	1 if malnutrition
		0 otherwise
	sjogrens	1 if Sjogrens syndrome
		0 otherwise
Skin Prick Test	antal3	number of negative reactions (0-3) (Result of skin provocation test to assess the patient inflammatory reactivity measuring the number of negative reactions in the patient.)

Table 2: Description of local predictors and outcome variable

Variable	**Variable name**	**Description**
Bleeding	bleed	1 if bleeding
		0 otherwise
Mobility	mobil	1 if mobility
		0 otherwise
Probing Plaque	plaque	1 if plaque
		0 otherwise
Endodontic factors	dtedont	0 if endodontically intact (a tooth that has its natural inner tissue (called pulp and nerve) and has not been treated with a procedure that removes this tissue and replaces it with synthetic material) or Root filled and periapically intact (a tooth that has undergone a procedure to remove its natural pulp and nerve tissue and fill it with synthetic material. The area around the root tip of the tooth shows no signs of bone loss or damage).
		1 if periapical destruction (when there is damage or loss of bone around the tip of the tooth's root) or root filled (natural inner tissue has been removed and replaced with synthetic material) but periapical destruction.
Pocket depth	depth	mm, average of the two surfaces
Restored surface	restorat	0 if no restoration
		1 if restoration in crown only
		2 if restoration is extending into the root on any of the two surfaces
		3 if overhanging restoration on any of the two surfaces
Initial bone level	bonelevel	mm, average of the two surfaces
Abutment	brotand	1 if abutment (a piece connecting a dental implant and an artificial tooth).
		0 if otherwise
Furcation involvement	furcat	1 if none of the surfaces has furcation
		2 if diagnostable <2mm on any of the two surfaces but none >2mm
		3 if >2mm on any of the two surfaces
Vertical destruction	destr	1 if vertical destruction on any of the two surfaces
		0 otherwise
Periodontitis indicator variable	progression0to2	0 if no indicators of periodontitis progression*
		1 if one indicator of periodontitis progression*
		2 if two or 3 indicators of periodontitis progression*
		*An indicator of periodontitis progression is a combination of radio- graphic marginal bone loss over time, development of clinical furcation involvement, and angular bony destruction. Periodontitis was considered to have progressed if one or more of the three peri- odontitis progression indicators had developed. The first indicator was: at any proximal surface (radiographic marginal boneloss, furcation involvement, or angular bony destruction), second was: at any proximal, facial, or oral surface (furcation involvement) and the third was: increased in severity (radiographic marginal bone loss or furcation involvement) between baseline examination and follow-up.

Table 3: Description of the three models

	Model 1	**Model 2**	**Model 3**
Systemic Variables	Age	Age	Age
	Family History	Family History	Family History
	Socioeconomic factors	Socioeconomic factors	Socioeconomic factors
	Patient's disease awareness	Patient's disease awareness	Patient's disease awareness
	Current Smoking	Current Smoking	Current Smoking
	Dentist's experience	Dentist's experience	Dentist's experience
	Systemic diseases	Systemic diseases	Systemic diseases
	Skin prick test	Skin prick test	
	Awareness and interest of disease after information		
Local Variables	Bleeding	Bleeding	Bleeding
	Mobility	Mobility	Mobility
	Probing plaque	Probing plaque	
	Endodontic factors	Endodontic factors	
	Pocket depth	Pocket depth	
	Restored surface	Restored surface	
	Initial bone level	Initial bone level	
	Abutment	Abutment	
	Furcation Involvement		
	Vertical destruction		

Table 4, Table 5 and Table 6 show descriptive statistics for the predictor and outcome variable. Table 4 and Table 5 contain the categorical variables while Table 6 contains the numeric variables.

Some of the predictors from Table 5 have only taken one value for all observations and the missing values are either zero or has a proportion 0.1408 to all of the observations. This is because of all the dropouts in the study as these values where collected at the end of the study, hence all the dropout observations are missing.

The mean age for the patients in the study shown in Table 6 was 47.9 years and the age span ranges from 29.7 years to 76.4 years with no missing values.

Table 4: Proportion of each category and missing values with a total of 3408 observations

	Category			
	0	1	2	Missing values
progression_0to2	0.3415	0.3278	0.0599	0.2708
plaque	0.3093	0.5508		0.1400
dtedont	0.3283	0.0114		0.6602
furcat	0.2180	0.0176	0.0261	0.7383
destr	0.6379	0.0833		0.2788
bleed	0.3178	0.5449		0.1373
restorat	0.3219	0.3216	0.1925	0.1640
mobil	0.4924	0.0552		0.4525
brotand	0.6922	0.0379		0.2700

Table 5: Proportion of each category and missing values with a total of 214 observations

	Category				
	0	1	2	3	Missing values
DTMOTHER	0.2488	0.1831	0.4272		0.1408
DTFATHER	0.2723	0.1408	0.4460		0.1408
DEPREG	1.000	0.000			0.000
diabetes_notcontr	1.000	0.000			0.000
diabetes_contr	0.9859	0.0141			0.000
blood	1.000	0.000			0.000
monogen	1.000	0.000			0.000
granulom	1.000	0.000			0.000
drug	0.9953	0.0047			0.000
osteop	1.000	0.000			0.000
malnutr	1.000	0.000			0.000
sjogrens	1.000	0.000			0.000
antal3	0.1737	0.1925	0.1831	0.4507	0.000
DTINTDIS	0.0188	0.2347	0.6056		0.1408
DTSOCEC	0.6385	0.2066	0.0141		0.1408
SKHAB	0.6948	0.1643	0.1174	0.0235	0.000
opexp	0.6056	0.3944			0.000

Table 6: Descriptive statistics for numeric predictors where *bonelevel* and *depth* has 3408 observations and *age* has 214 observations

	Mean	Sd	Min	Max	Missing values
bonelevel	3.2449	1.6559	0.75	12	0.2139
depth	3.7882	1.0120	3	11	0.2650
age	47.9142	10.3769	29.7096	76.3507	0.000

2.2 Missing values

The original data set contains missing values, where the number of missing values in each variable varies but consisted of a reasonable amount of missing values to still be included in the data set. To deal with the missing values we implemented a missing values imputation function in R, that is using Random Forest proximities. More about this will be explained in Theory & Method.

Random Forest requires that the outcome variable does not include any missing values. Therefore, all observations with missing values in the outcome variable are removed from the analysis. This resulted in a final data set of 2485 observations. These missing values are because of patients that have dropped out of the study.

3 Theory & method

3.1 Tree Based Methods

Tree based models are supervised machine learning methods that are widely used to make predictions. Supervised in the context of machine learning means that we have known input observations and also have information regarding the output. We assume that there is a meaningful structure from the input predictors that we want to use to group our observation. The tree is a hierarchical structure which divides the input data into smaller regions until a prediction is assigned to each region. This can be done for both regression and classifications tasks (James et al., 2021). To illustrate this and to give an intuitive explanation for how this works we use an example.

3.1.1 Example: Predicting suitable weather for playing ball outside

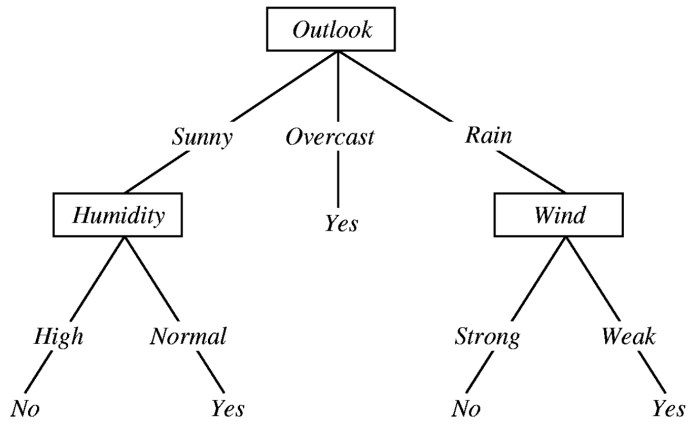

Figure 2: Structure for a classification decision tree made to predict whether the weather will be suitable for playing ball outside with the two classes "Yes" or "No". The picture is taken from Yadav, P (2023-05-03).

The decision tree from Figure 1 starts at the top with all observations in the same region which is the first node. Every node represents a feature. Then it divides the data into three regions, which are called branches, using a certain splitting rule which in this case is if the weather is "Sunny", "Overcast" or "Rain". For the observations that have been divided into the "Humidity" and "Wind" node, another split is done. The split is done based on if the "Humidity" is "High" or "Low" and for the "Wind" node if the wind is "Strong" or "Weak". If the weather is "Overcast" the prediction is "Yes" without any more splitting to that data. If the features instead would be continuous, the splits are done based on numeric thresholds

for the feature. The final predictions ("Yes" and "No") are called the terminal nodes or leaves. All the observations are passed down the tree and are to be predicted as "Yes" or "No" when doing classification and for regression a continuous value will be predicted.

Tree based models such as in this example are simple to understand and easy to interpret but are not very competitive compared to other more complex approaches to machine learning. Therefore, ensemble methods for decision trees such as Bagging, Boosting, and Random Forests have been developed and are widely used (James et al., 2021).

Bagging uses bootstrap to create subsets on the original data where the observations are selected with replacement, a decision tree is then created for every subset which are independent of each other. We get the final predictions by merging all the predictions from all the trees from all the subsets. Meanwhile, Random Forest is an extension of Bagging which also selects subsets of variables used in each subset to divide the data sample and create a decision tree (Cutler et al., 2012).

3.2 Random Forest

The person that introduced Random Forest was Leo Breiman. The method was meant to compete with boosting and it came up as an addition to Breiman's bagging idea. Random Forests has the ability to both classify and make regression. In classification the response variable is categorical, while in regression the response variable is continuous. The same implies for the predictor variables, where they can be both categorical and continuous. (Cutler et al., 2012).
Random Forest is a method that requires low computational power and is easy to implement. Despite these characteristics it is praised as of the representatives of the modern ensemble learning methods and often generates good results when using it on real world problems (Zhou, 2021). Implication can be made to a large variety of tasks, such as predicting vegetation type occurence, predicting distributions of characterisitcs in animal species, diagnosing diseases and more (Verikas et al., 2011).

The way Random Forest work is that it bootstraps a certain number of samples from the training data and then create decision trees on them. In each split in the decision tree, the model provides a random sample of m predictors from the total number of predictors p. From

this random sample of predictors the split can only choose one of them to make the split. The purpose of the extension that Random Forest provides is that it prevents the bagged trees from becoming correlated, which traditional Bagging has a tendency to make. For example, if a certain predictor in the data is stronger than the rest of the predictors, this specific predictor will be used in the first split of every tree. This is not a good thing since the trees will look very similar to each other and high correlation between them will not reduce a large quantity of the variance. The random sample of predictors that Random Forest provides makes the trees decorrelated and reduces more of the variance (James et al., 2021).

Random Forest has some statistical advantages, such as it is immune to statistical assumptions and preprocessing burden. Furthermore it controls overfitting and detects outliers. It also has computational advantages since it can perform well despite noise and high dimensionality which makes it a very robust method (Zhu, 2020, August).

The choice of doing predictions with Random Forest was mainly done because of its robustness even though other methods where considered. The other methods looked into was for example Logistic regression, the machine learning method Boosting and Neural Networks. But as this thesis includes predicting to more than binary outcomes and as Random Forest is not very dependent on any assumptions, Random Forest was considered as the best suitable choice for the data available and for the specific purpose.

3.2.1 Out of bag estimated error

Not all observations are included in each of the bootstrap samples when constructing the trees. These are called out-of-bag (OOB) observations and can be used to obtain the prediction performance of a model. Since each tree does not include all the observations, each observation will not be included in some of the trees, they are out-of-bag. By using the trees where a certain observation is not included, the prediction for this specific observation can be obtained. By doing this for all the observations included in the Random Forest process, every observation get a classification and an error rate can be estimated (Janitza & Hornung, 2018).
This error rate will be compared to the error rate obtained from the test set that is left out when training the model. If the difference is small, the model can generalize good to data it was not trained on.

3.2.2 Missing values Imputation

Briefly the proximity in Random Forest can be described as the proportion of the time two observations end up in the same terminal node. The proximity can take values between 1 and 0, where 1 means they always end up in the same terminal node and 0 means the opposite. The proximity is used in the missing values imputation, which is an iterative procedure that first uses median imputation to all the missing values. Based on that new data set and the proximities it then makes predictions for all the original missing values. The predictions of the missing values are then used to complete the data without any missing values. (Cutler et al., 2012).

3.3 Tuning

Random Forest is, compared to many other machine learning methods, known to be working considerably well without much tuning to the parameters. Although, there are three parameters which can be tuned and in some situations give the model a better performance. These are:

- m, the number of randomly selected predictor variables chosen for each node
- J, the total amount of trees in the Random Forest
- *tree size*, how big the tree is, it is often measured as the maximum of terminal nodes that we allow.

Even though it is possible to tune all three parameters, Random Forest is only (in some situations) sensitive to m. When doing classification the standard default is to set $m = \sqrt{M}$, where M is the number of predictors. However, if found that fine-tuning of the parameter is needed, m is chosen by looking at the out-of-bag error rate. Although doing so means that the estimation of the generalization error is no longer unbiased. Usually, this is not needed as Random Forests in most cases are not very sensitive to m. Hence, fine-tuning is often not needed and the risk for overfitting due to the choice of m is relatively small. (Cutler et al., 2012)

An often occurring problem when working with ensemble methods is that the models tend to

be overfitted when J becomes large, which also is associated with an increase in the generalization error. However, this is not the case for Random Forests. The only problem regarding J in Random Forest is when J is too small. Empirically, the forest will get the smallest errors when the number of trees grows to infinity but the computational time will take too long (for a normal computer) when J does so. Therefore, the out-of-bag error rate is used to decide when J is large enough, which is when the estimated generalization error has been stabilized. (Scornet, 2017)

The *tree size* can sometimes be a problem if it is too big, meaning that there are too many terminal nodes. This can lead to overfitting of the model and to cope with this the out-of-bag error rate can once again be used for tuning the parameter. Although, this will lead to bias in the estimated generalization error and is in most cases not necessary. (Cutler et al., 2012)

Considering this, $\sqrt{M}$ will be used as the number of predictors in each split and use 500 trees since that is the default setting and a fair amount to obtain good results.

3.4 Evaluation of the models

3.4.1 Variable Importance

It is often crucial when doing a classification or regression prediction to estimate which of the variables used for the prediction that have the biggest impact. This is important both for interpretation of the fitted forest and also for variable selection to see if the forest can be refitted with only the variables that were found to have a big variable importance for the analysis. To find the Variable Importance we use the measure of Permutation Importance (Cutler et al., 2012).

3.4.2 Permutation Importance

To measure the importance of a specific variable k, Permutation Importance is made. This is done by using the out-of-bag data for each iteration of trees and then the Random Forest model that is estimated is used to predict the out-of-bag data into classes. From those predictions, the prediction error for the out-of-bag sample is calculated. To find if the variable has some

importance in predicting, the same out-of-bag data sample is used but with the values for the variable of interest k. k is then randomly permuted in the out-of-bag data sample while keeping all other variable the same. This modified out-of-bag data is then passed down in the Random Forest model to predict the classes for the modified out-of-bag data sample and the prediction error is calculated. The two prediction errors are compared and if the variable has some importance, the unmodified out-of-bag prediction error should be lower than the modified out-of-bag data sample. The difference between the prediction errors are a measure of the Variable Importance and the overall Variable Importance is computed by taking the average from all the observations. In regression, the same method is applied but instead of using the difference between the prediction errors, mean squared error (MSE) is used (Cutler et al 2012).

One way to measure the variable importance is to use Mean Decrease Accuracy. This method randomly permutes the values of the variables and then measures the prediction accuracy change. Then it obtains the Variable Importance for each variable (Behnamian et al., 2017). This measure will be used to investigate the Variable Importance in this report.

3.4.3 Confusion Matrix

A common way to evaluate a machine learning model is to visualize the model's predictive performance in a confusion matrix. The matrix shows all the predicted classes as well as the true/actual classes combined in a matrix. Hence, it shows the number of true positive (TP), true negative (TN), false positive (FP), and false negative (FN) predictions made by the model on the validation data set, as shown in Table 7. The total amount of positive predicted are denoted P and the negative predicted values are denoted N. P is calculated separately for Class 1 and 2. These are used to calculate the overall accuracy (formula (1)), sensitivity (formula (2)), specificity (formula (3)), positive predicted value (formula (4)) and negative predicted value (formula (5)). Note that these measures are typically for binary classification although it still works for multiple classification problems (James et al., 2021).

In this case with three different classes it gets more complicated. To calculate these metrics for multiclass classification we have defined Class 0 as the negative and Class 1 and 2 as positives. The measures are then calculated separately for each class. For example when calculating the sensitivity for Class 1 where formula (2) is used, TP, FN & FP for Class 1 is considered. Hence only the middle column in Table 7 is considered in the calculations as

we are interested in the actual values. If Positive predicted value for the same Class would be calculated from formula (4) only the row for Class 1 would be used. The same implies for Class 2. For Class 0 classification is measured in terms of specificity and negative predicted value instead, since this Class contains the patients with no risk indicators of Periodontitis.

Table 7: Confusion matrix for three classes

		Actual Class			
		0	1	2	Total
Predicted Class	0	TN	FN	FN	Total no. with no predicted disease
	1	FP	TP	FP	Total no. with predicted disease Class 1
	2	FP	FP	TP	Total no. with predicted disease Class 2
		Total no. negatives	Total no. positives Class 1	Total no. positives Class 2	Total no. of observations

3.4.4 Evaluation measures

The overall accuracy represents the percentage of correctly predicted classes:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

The Sensitivity represents the proportion of the true positives that were classified correctly for each class:

$$\text{Sensitivity} = \frac{TP}{TP + FN + FP} \tag{2}$$

Specificity represents the proportion of true negatives that where classified correctly for each class:

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{3}$$

Positive Predicted Value represents the proportion of predicted positives that where a true positive:

$$\text{Positive Predicted Value} = \frac{TP}{P} \tag{4}$$

Negative Predicted Value represents the proportion of predicted negatives that where true negatives:

$$\text{Negative Predicted Value} = \frac{TN}{N} \tag{5}$$

The overall accuracy is a evaluation measure of the performance for the whole model meanwhile sensitivity, specificity, positive predicted value and negative predicted value are measures of the performance for the model for each class separately. Depending on the purpose of a model, different measures can be more or less important (James et al., 2021).

3.5 Software

For all calculations when estimating the Random Forests together with their corresponding results and the variable importance estimations, the open source statistical software R has been used.

4 Results

4.1 Accuracy & error

Table 8 shows the overall accuracy for the training- and validation set, the error for the validation set (1-accuracy), out-of-bag (OOB) estimated error and the difference between the OOB estimated error and validation set error. For Model 1 the accuracy for the training set was 0.9701 and the validation set 0.6823. The error for the validation set was 0.3177 and the OOB estimated error was 0.3232. Therefore, the difference in error was 0.0055.

Model 2 had a similar difference in accuracy for the training and validation set as Model 1, where the accuracies were 0.9051 respectively 0.5965. Further, the errors were larger than in Model 1 and the error difference was larger.

Model 3 with the smallest set of variables made the worst predictions with a training set accuracy of 0.6141 and a validation set accuracy of 0.5322.

Table 8: Accuracy and error measures for all models

	Model 1	Model 2	Model 3
Accuracy training set	0.9701	0.9051	0.6141
Accuracy validation set	0.6823	0.5965	0.5322
Error validation set	0.3177	0.4035	0.4678
OOB estimated error	0.3232	0.3790	0.4324
Error difference	0.0055	0.0245	0.0354

4.2 Confusion matrix & evaluation measures

Table 9 shows that Model 1 never predicted Class 2 when the actual was Class 0. The same applies when the model predicted Class 0 and the actual was Class 2. As shown in Table 10 the specificity (proportion of teeth with no disease that were correctly predicted) for Model 1 was 0.7799 which was higher than both sensitivity measures (proportion of teeth in a certain disease class that were correctly predicted), for Class 1 with a value of 0.6032 and Class 2 with a value of 0.5079. Regarding the positive predicted values the model performed best with Class 2 where 0.8205 of the teeth predicted to disease class 2 actually belonged to Class 2, compared to 0.6291 for Class 1. The negative predicted value (proportion of teeth predicted to no disease, that actually had no disease) was 0.7086.

Table 9: Confusion Matrix for Model 1. The colored cells are the correctly predicted classes

		ACTUAL		
		Class 0	Class 1	Class 2
PREDICTED	Class 0	**287**	118	0
	Class 1	81	**190**	31
	Class 2	0	7	**32**

Table 10: Sensitivity and specificity measures for each class for Model 1

	Class 0	Class 1	Class 2
Sensitivity		0.6032	0.5079
Specificity	0.7799		
Positive Predicted Value		0.6291	0.8205
Negative Predicted Value	0.7086		

The confusion Matrix in Table 11 shows that compared to the previous model's Class 2, one observation is predicted to Class 2 even though the actual class was Class 0 and 9 observations where predicted to Class 0 even though the actual class was Class 2. Table 12 shows that the specificity for Model 2 of 0.6676 was higher than both the sensitivities for Class 1 of 0.5913 and for Class 2 which was 0.2097. A similar pattern is shown regarding Model 2's positive predicted values and the negative predicted values. The negative predicted value was 0.6531,

and the positive predicted value for both Class 1 and 2 was both lower than 0.6531. The positive predicted value for Class 1 was 0.5457, meanwhile it was even lower for Class 2 with a value of 0.4815.

Table 11: Confusion Matrix for Model 2. The colored cells are the correctly predicted classes

PREDICTED \ ACTUAL	Class 0	Class 1	Class 2
Class 0	**241**	119	9
Class 1	119	**191**	40
Class 2	1	13	**13**

Table 12: Sensitivity and specificity measures for each class for Model 2

	Class 0	Class 1	Class 2
Sensitivity		0.5913	0.2097
Specificity	0.6676		
Positive Predicted Value		0.5457	0.4815
Negative predicted value	0.6531		

From Table 13 we see that there are zero correctly predicted observations in Class 2 for Model 3. This results in a sensitivity and a positive predicted value of zero for Class 2. Therefore, Model 3 is even worse at predicting for Class 2 than the previous model. According to Table 14 the specificity is clearly higher than the sensitivity for Class 1 meanwhile the negative predicted value and the positive predicted value for Class 1 are similar, both around 0.5.

Table 13: Confusion Matrix for Model 3.
The colored cells are the correctly predicted classes

PREDICTED \ ACTUAL	Class 0	Class 1	Class 2
Class 0	**258**	212	14
Class 1	82	**139**	38
Class 2	0	3	**0**

Table 14: Sensitivity and specificity measures for each class for Model 3

	Class 0	Class 1	Class 2
Sensitivity		0.3927	0.0000
Specificity	0.7588		
Positive Predicted Value		0.5367	0.0000
Negative Predicted Value	0.5331		

4.3 Variable importance

Figure 3 shows the Variable Importance for Model 1. The predictor with the highest importance was *destr* and the second highest was *furcat* which both had mean decrease accuracies above 40, followed by *age* and *bonelevel* which both were around 30 in mean decrease accuracy. The other predictors were all low in importance with a similar level of importance. Just the ten most important variables were included in the plot which gives an overall understanding of the variables' importance.

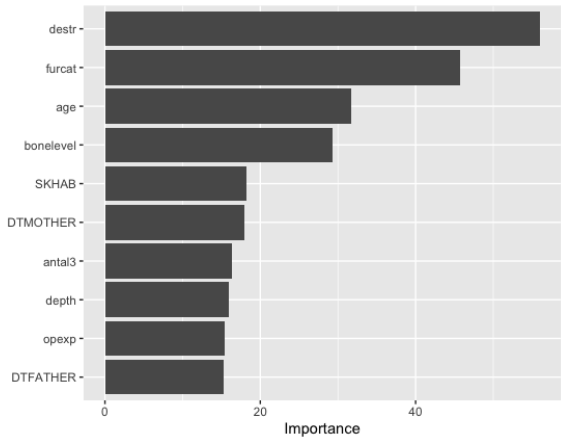

Figure 3: Variable Importance Plot for Model 1, showing the indicators with the highest importance and excluding the indicators with little to no importance.

According to the Variable Importance for Model 2 in Figure 4, *age* had the most importance in the prediction with a mean decrease accuracy of approximately 42, followed by *bonelevel*, *antal3* and *depth*.

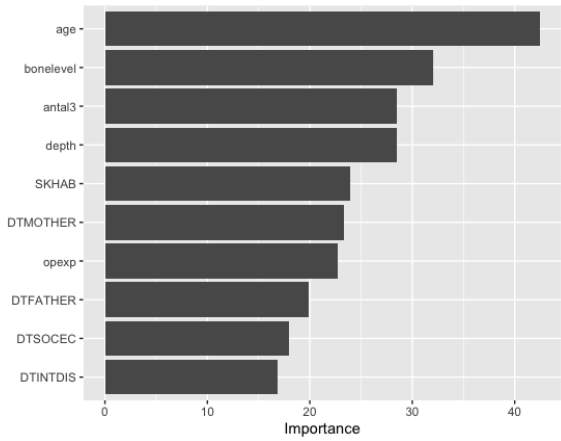

Figure 4: Variable Importance Plot for Model 2, showing the indicators with the highest importance and excluding the indicators with little to no importance.

According to the Variable Importance for Model 3 in Figure 5, *age* had the most importance in the prediction with mean decrease accuracy of approximately 35, *opexp* was second, *DTMOTHER* and *SKHAB* was third and fourth closely followed by *mobil*, *DTFATHER*, *diabetes_contr* and *DTINDIS*. Note that compared to Models 1 and 2 none of the variables had an importance above 40 in mean decrease accuracy.

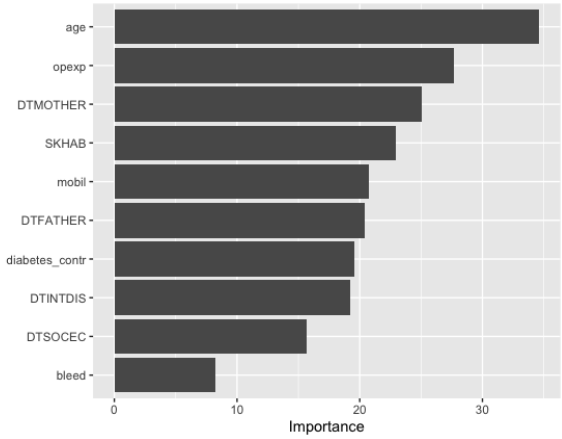

Figure 5: Variable Importance Plot for Model 3, showing the indicators with the highest importance and excluding the indicators with little to no importance.

5 Discussion

The focus of this report was to make predictions with an outcome variable that can obtain three different values. It was found that Model 1 with all predictors available included, had the best performance in terms of accuracy and error estimation. The error difference between the OOB and validation set was 0.0055 which means the model can estimate the error on new data well. One thing that seemed suspicious was that the differences between the accuracy in the training set and the validation set in Model 1 and Model 2 were big relative to the difference in Model 3. Hence, there is cause to believe that some level of overfitting is made, although after doing research and testing with different tuning of the models no solution for this was found. And as Random Forest is very robust to overfitting we believe that this is not a major issue for the models' ability in doing predictions. Although, there is obviously something that affects the models' ability to make predictions on unseen data.
Model 3 made the weakest predictions and had the largest difference in error estimation, therefore Model 1 followed by Model 2 had the best performance.

From the evaluation measures, Model 2 and especially Model 3 were found to be not very good at doing predictions for Class 2. It might be the case that the models can not separate the positive outcomes (Class 1 and 2) very well. On the other hand, all the models had quite similar values for specificity and negative predicted value which means that it was easier for the models to find the patients that do not have Periodontitis and exclude the patients that have progression of Periodontitis.

Some of the predictors with the highest importance turned out to be *destr*, *furcat*, *age* and *bonelevel* and as *destr* and *furcat* only was used in Model 1 that might be a possible explanation to why it performed the best. Comparing the predictors which had high importance in Model 3 to their respective importance in Model 1, it is clear that they are not good for doing predictions as their importance is low compared to the best predictors which were used in Model 1. As Model 3 was based only on systemic predictors and for both Model 1 and Model 2 the local predictors are over represented in having high importance, it might be that the local predictors were more useful in doing predictions for our data.

5.1 Limitations

A common phenomenon regarding the topic of teeth and mouth care is people's fear of the dentist. There is a credible risk that people who are afraid or do not like going to the dentist avoid going there. Therefore, they will not be included in the study and we miss out on a certain group of the population. We do not know if this group tends to take care of their teeth and health more or less compared to the general population, or if Periodontitis is more frequent within that group. Hence, there might be some level of bias in the study.

A similar problem regarding bias exists as there are patients in the study who did not go through with the study till the end, resulting in that they did not provide any final value for having some type of Periodontitis or not. These observations are therefore excluded from the study and if the patients who dropped out of the study are not completely random, this might also lead to bias.

5.2 Further studies

As there were indications of problems regarding predicting for Class 2, it could be of value to investigate further how a model with only a binary outcome variable would perform. A binary outcome would eliminate the problem of not being able to separate the positive outcomes. If such an approach is done, Random Forest might not be the most suitable method for this purpose and other methods such as logistic regression, linear discriminant analysis, neural networks or other machine learning techniques should be considered. Binary classification is also very suitable for evaluations using ROC-curves which could have expanded the evaluation between the models.

As the study was based on if a patient will develop some indication of Periodontitis progression after 4 years, it would also be interesting to see how the risk indicators would affect the risk of Periodontitis not only within 4 years but also for a longer time period.

6 Conclusion

The purpose of this report was to create an algorithm with Random Forest that can predict the risk of someone developing Periodontitis. Of the three models we tried, Model 1 performed the best. It had an overall accuracy of 68.23% which may not be considered high. It was also found that the local predictors had a higher importance for the predictions which is why Model 1 performed the best, as it had the most of these included in the algorithm. Although, the model still gives some indication of the risk level for progression of Periodontitis. Therefore, the predictive algorithm should be used as a supplementary tool in conjunction with expert opinions for assessing the risk of Periodontitis progression.

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